NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

# R23.3b AI-hybrid synthetic typhoons to coastal design wind speeds

**Group** AI Weather & Climate Models for Wind Engineering

**Status** Shelved

**Period** 2026-10-14 19:18 — 2026-10-16 00:57

## Stage conclusion

## Project report

# B2 AI-hybrid synthetic typhoons to coastal design wind speeds

## Objective

Use an AI/ML-assisted synthetic tropical cyclone catalog (RAFT: statistical tracks + MLP intensity) coupled with the parameterized wind field that passed Gate (a) in B1+B4 (CLIMADA 6.1.0 H08 + roughness) to compute 50- and 700-year return-period wind speeds at US Gulf and Atlantic coastal stations; compare these against the traditional statistical catalog STORM run through the same wind field chain and against ASCE 7-22 design wind speeds. Determine whether "AI synthetic typhoons → design wind speeds" is worth continuing (replacing the wind field model once Liang Hu's model is ready). Task brief.

# Criteria (frozen 10-14 19:25 after both EVA and Design Loads tracks replied; no post-hoc relaxation permitted)

**Unified convention**: Station height 10 m, open terrain (z0 = 0.03 m, corresponding to ASCE Exposure C), 3 s gust, denoted V50 and V700 (MRI 50 and 700 years). Basis: Vickery et al. 2010 give 300/700/1,700-year maps as peak gusts at 10 m in open terrain (z0=0.03 m), where Category II is 700 years (card, PDF pp. 1, 5–8 / journal pp. 613, 617–620; that paper does not explicitly state "3 s"); "3 s" is taken from ASCE 7-22 Chapter 26, where the basic wind speed V in ASCE 7-22 is the nominal 3 s gust at 10 m (33 ft) in Exposure C (close reading line 23, Ch. 26; exact page number mapping Risk Category II to 700 years pending verification against the original text by ideas). All stations are uniformly computed assuming open terrain, without directional station-exposure corrections (the ASCE maps themselves assume uniform Exposure C).

- **Gate 0 (wind field chain qualification)**: Across stations, the median bias of V700 from the STORM chain relative to ASCE 7-22 must be within ±20%. If it fails → record the conclusion as "wind field / conversion chain unqualified" and do not evaluate AI (same logic as Gate (a) in B1+B4).
- **Gate 1 (worth continuing)**: Across stations, the median absolute bias of V700 from the RAFT chain relative to ASCE 7-22 must be ≤ 15%, **and** ≤ 1.1 × the median absolute bias of the STORM chain. If both are satisfied → worth continuing; otherwise shelve and clearly document where the discrepancy lies (catalog intensity/frequency, wind field model, or conversion convention).
- **Report-only, non-gating**: Three-way V50 comparison; breakdown by coastal segment (Texas–Louisiana / Mississippi–Florida Panhandle / Florida Peninsula & Keys / Georgia–North Carolina); sampling uncertainty of return periods (annual-resampling bootstrap); sensitivity to averaging-time / gust conversions and the STORM 10 min→1 min conversion; if a public SPIN or Bose 2023 catalog is found, include it as a third chain for reporting only.
- **Decision rule if ASCE values are unavailable (pre-specified at freeze)**: ASCE 7-22 station values are retrieved strictly from public sources (ATC Hazards by Location, etc., without using Liang Hu's credentials; no reading off code PDF maps to make up numbers). If ≥12 out of 19 stations are obtained → compute and judge Gate 0 and Gate 1 on the obtained stations as usual. If <12 stations are obtained → do not judge Gate 0 / Gate 1; downgrade to a **relative comparison**: if the across-station median absolute difference in V700 of RAFT relative to STORM is ≤15%, and the median difference in each of the four coastal segments is within ±25%, record "RAFT is consistent with the traditional catalog and may continue (untested against code)"; otherwise record "RAFT is inconsistent with the traditional catalog" and specify where the discrepancy lies. Conclusions from a relative comparison must not be stated as equivalent to design wind speeds.
- **Design Loads track materials are read-only** (10-14 reply from Design Loads Lead): May cite the ASCE 7-22 wind interface close reading (needs_review), (PDF page = print page + 61), and the code text itself. Do not modify their calculators; do not cite or use for endorsement the numerical artifacts named "B2/C0" on their track (those belong to their own line, not this node).
- **Stratified reporting (added by parent project on 10-14, report-only, non-gating)**: H08 is known to overestimate across broad outer regions by ~+2.5 m/s (Node H) and underestimate in the inner core <1 RMW. For each station's V50 and V700, report the breakdown of exceedance contributions by storm track distance to the station in bins of <1, 1–3, and >3 × RMW (classified by distance/RMW at the time of

peak station wind among storms exceeding that return-period wind speed); additionally, stratify and report station biases by distance to the coast (<10, 10–50, >50 km).
- Constants, stations, and methods must be written into (timestamped) before computing any station wind speeds; no changes afterward—any required modification must be logged as v2 while preserving v1.

## Stations (frozen 10-14 19:25, 19 ASOS airport stations from Texas to North Carolina)

KBRO Brownsville, KCRP Corpus Christi, KGLS Galveston, KLCH Lake Charles, KMSY New Orleans, KGPT Gulfport, KMOB Mobile, KPNS Pensacola, KAAF Apalachicola, KTPA Tampa, KRSW Fort Myers, KEYW Key West, KMIA Miami, KPBI West Palm Beach, KMLB Melbourne, KJAX Jacksonville, KCHS Charleston, KILM Wilmington, KHSE Cape Hatteras. Only station coordinates are used (IEM station table); ASOS wind observations are not needed. If an ASCE value cannot be retrieved for a station, record it as missing and do not substitute the station.

## Current Status

**10-14 22:30 Shelved** (6/6 in favor; verdict). - **Result** (primary version v6, RAFT annual rate using HURDAT2 observational proxy of 12.075 storms/yr): Gate 0 passed (STORM V700 median bias vs. ASCE 7-22 is −8.87%, 19/19 stations); Gate 1 failed (RAFT median absolute bias is 30.95%, >15% and >1.1×9.21%). RAFT is biased low across all four coastal segments; V50 is also biased low by −29%, showing an overall distributional low bias rather than tail-sampling noise. Sensitivity tests across annual rates of 10–20, POCI ±5 hPa, and conversion factors of 1.03–1.12 do not alter the verdict. - **Why it failed**: The shared wind field chain is controlled by STORM passing Gate 0, and units/inputs/plotting were verified against CLIMADA 6.1.0 source code by 4 reviewers, finding no implementation error capable of explaining a 30% discrepancy. Because H08 uses only pressure deficit and does not read catalog vmax, this chain actually tests **the empirical Pc/Rmax formulas supplied by RAFT**, not its AI intensity model per se. Whether the low bias stems from the empirical Pc–vmax relation (intercept 1030 hPa paired with POCI 1010), downward-biased Rmax regression, nearshore decay, or shared assumptions was **not quantitatively separated** in this round; the authors' self-reported biases (Rmax ~19% low, landfall 99th percentile ~8 kt low, card NR_xu_2024_raft PDF pp. 6–7) serve only as directional clues. - **What is missing**: A wind field model that directly ingests vmax (or a self-consistent Pc–vmax relationship), plus a distributional comparison of RAFT Pc/Rmax/nearshore intensity against HURDAT2/EBTRK. - **Conditions for restart**: Prerequisite work—either Liang Hu's wind field model becomes available, or a distributional diagnostic of RAFT Pc/Rmax/nearshore intensity against HURDAT2/EBTRK is completed first (initial hypothesis: pairing the linear Pc formula with purely pressure-driven H08 loses ~10–15% at the strong-typhoon end, estimated by reviewer E, not yet recomputed); **Trigger**—write a new pre-registration based on those findings and re-run under the exact same frozen thresholds; restart succeeds only if Gate 0 still passes and Gate 1 passes, without tuning any parameters against this round's ASCE errors. If a new public AI synthetic catalog emerges that natively includes Pc/Rmax (which SPIN currently lacks), it may also be tested under the same

protocol. - **10-15 00:10 B2 wrap-up (status remains Shelved)**: Attribution complete—the primary cause is that the RAFT catalog itself produces almost no Cat3+ passages nearshore (especially in the Gulf of Mexico and the Carolinas), rather than the empirical Pc/RMW formulas or the wind field model. Consequently, switching to a wind field model that directly uses vmax (H10) cannot rescue it, because vmax itself is too weak nearshore. **Revised restart condition**: Only when a new AI synthetic catalog (or a new version of RAFT) becomes available, first run a low-cost screen (0 SU) using this diagnostic's "near-station Cat3+ passage rate vs. HURDAT2" table; only if passage rates across segments are of the same order of magnitude as observations should it enter the design wind speed chain, with a fresh pre-registration under the original frozen thresholds and 19 stations. Local copies of raw public data have been deleted from the workstation (re-fetchable). - **10-14 23:30 Diagnostic v1 returned**: Neither the empirical Pc formula nor RMW is the primary cause, leaving no pre-registerable correction, so no restart PREREG will be drafted; the remaining discrepancy lies in "intensity × passage frequency of strong near-station storms," and ideas has been dispatched for the final step: near-station strong-storm passage frequency of RAFT/STORM vs. HURDAT2 (artifact) to wrap up B2. - **10-14 22:45 Pre-restart diagnostic dispatched** (Lab Lead, status remains Shelved): ideas (queued), comparing vmax/Pc/ΔP/Rmax of RAFT/STORM against HURDAT2+EBTRK, swapping Pc and Rmax separately in the same H08 chain to inspect single-storm station peak wind changes, and checking CLIMADA wind field models that directly use vmax; without touching ASCE or recomputing return periods; task brief, artifact. Upon return, decide whether to draft a restart PREREG (report drafting to Node B, report to Lab Lead before running gates). - Report errata (incorrect source for 14.5 storms/yr, relative comparison is not a gate, two undisclosed verification checks).

Opened 10-14 19:20; both EVA and Design Loads tracks confirmed no overlap and no borrowable station values or catalogs, and criteria and stations were frozen at 19:25 (including relative comparison rules if ASCE values are unavailable). Round 1 dispatched to ideas (data fetching, PREREG, dual-chain calculation, ASCE value retrieval, GitHub repository table). Literature: Grok quick check available (clues only, unreviewed); Deep Research monthly quota exhausted, Step 2 to be supplemented once quota resets, followed by ` litwiki card creation based on its conclusions.

# Established Decisions

- 2026-10-14 22:30 Shelved (review 6/6 in favor) →
- 2026-10-14 19:25 Both tracks replied, criteria frozen (added relative comparison rules if ASCE values cannot be retrieved, plus read-only and prohibited items for Design Loads track materials) → "Supplement 2"
- 2026-10-14 Parent project added two boundaries: wait for replies from EVA / Design Loads tracks before freezing criteria; add stratified reporting by RMW distance and coastal distance → "Supplement"
- 2026-10-14 B2 kickoff, draft criteria and stations, Gate 0 prerequisite →
- 2026-10-14 B2 opened (Node B Lead) →

# Boundaries

- **Workstation does not run computations or store compute data (tightened by Liang Hu on 10-14, recorded 10-15)**: All wind field / catalog diagnostics, batch runs, and large downloads must be executed and stored on Colab (CPU/GPU) or CRC/TACC; the workstation only holds scripts, small results <200 MB per item, figures, and reports; if temporary storage >1 GB is strictly necessary, report to Node B first. Public raw data (large RAFT/STORM/HURDAT2 files, GFS/ERA5 fields) belong in Colab sessions or TACC [local path omitted] Supercomputer usage follows ComputeOps resource request + registration, with pre-check and submission separated into two commands; CRC defaults to single-node with group storage at ~95%, so large datasets prioritize TACC. Task briefs upon any restart must follow this rule; "workstation local priority" in older entries below is void.
- No single-experiment cap on Colab, but pause and report to Node B Lead if projected usage is >100 CU; run `colab stop` when finished; dispatch only one executor per round. Do not download RAFT rainfall files—only track `.nc` files; do not store >5 GB on the workstation.
- Do not use supercomputers. Check literature before making assumptions; the parameter source table is below. Record new data/code, and annotate data sources on all figures and tables.
- **EVA track reply (10-14, forwarded via parent project)**: EVA covers only thunderstorm high winds and mechanism-stratified extreme value analysis, with no overlap or restricted zones relative to B2; it has no ASCE 7-22 station values or hurricane catalogs to lend (ASCE materials belong to the Design Loads track, reply pending). Two caveats: ① B2's hurricane gust factors / averaging-time conversions must not be directly applied to thunderstorm or mixed-climate EVA (EVA candidate entry); ② if B2 results are later combined with the thunderstorm side into mixed-climate design winds, notify the EVA track via the parent project first to align conventions, and final conclusions on mixed-climate synthesis and return periods rest with Liang Hu. Available for reference but likely unneeded: Cook ASOS four-station Texas gust event inventory (read-only; not stratified by tropical cyclones, gusts not standardized to 3 s), not used in B2.
- Tier-2 node: do not open Tier-3 nodes; tool bots must not delegate further. EVA / Design Loads tracks are read-only; if their materials are needed, report to Node B Lead to request them.

# What Was Tried and Results

| Date | What Was Done | One-Line Result |
|---|---|---|
| 10-14 23:54 | Passage frequency attribution (using observations only; + ADDENDUM) | Cat3+ passages within 100 km of stations (per 1,000 storms): TX segment RAFT 0.09 / STORM 1.04 / obs 3.45; LA–Panhandle 0.17 / 1.45 / 4.14; FL Peninsula 1.97 / 2.55 / 2.07; Carolinas 0.42 / 2.66 / 3.45. RAFT has plenty of near-station TS-strength passages, but very few reach Cat3+ at passage (about 20× lower in the Gulf), while the FL Peninsula is close to observations—matching the segment pattern of V700 low bias (deepest in the Gulf). Answer: (iii) both contribute, dominated by the nearshore intensity tail; observational Cat3+ sample is small, so compare order of magnitude only |
| 10-14 23:20 | Pre-restart diagnostic v1 (comparing observational inputs only, without touching ASCE; paired replacement on a 5% sample of 2,351 storms in the same H08 chain, baseline verified item-by-item against the main chain) | Near-station Cat3+ ΔP p90 in RAFT is 71.6 hPa (obs 87, STORM 88.9), and RMW distribution is narrow (p99 52 vs. obs 150 nmi). **Replacing Pc only**: median change in affected station peaks is 0.0%, Cat4/5 +1.84%, refuting reviewer E's "10–15% loss" (observational static eyewall check median 2.87%); **Replacing RMW only**: median +4.2%, whereas applying the same replacement to STORM yields +16.7% and node-by-node sampling introduces jumps, precluding a fixed correction. Neither explains the ~30% gap. CLIMADA 6.1.0 H10 `vmax_from_cen=False` allows catalog vmax to enter the wind field (source lines 249–318, 503–508), not yet validated against observations. Currently no pre-registerable correction exists |
| 10-14 21:41 | Integrated page numbers into new litwiki cards (PREREG_v7 source addendum, post-results, no parameter changes or re-judging) | Correction: "STORM 14.5 storms/yr" written in v1 has no basis in the original paper; the STORM generation paper gives North Atlantic Poisson λ=10.8 (KHFQRECZ PDF p. 4); STORM is always evaluated using its own 10,000-year statistics, so gate verdicts are unaffected; 14.5 is retained strictly as a RAFT annual-rate sensitivity value. RAFT authors' self-reported low bias is used only as a directional clue, not quantitative attribution |

| Date | What Was Done | One-Line Result |
|---|---|---|
| 10-14 21:50 | Switched annual rate to HURDAT2 contemporaneous, matched-threshold observational proxy (1979–2018, TS/HU ≥35 kt and ≥24 h, 483 storms = 12.075 storms/yr → 3,313 equivalent years; PREREG_v5/v6, post-processing recomputation only, 0 CU) | Primary gating version: Gate 0 passed (−8.87%, unchanged); Gate 1 still failed (RAFT 30.95%, >15% and >1.1×9.21%); sensitivity rates of 10/14.5/20 storms/yr and unfiltered lifetime all yield 29–31%, leaving the verdict unchanged. Note: v6 was written to disk after the 14.5-version results (though the decision preceded the results), which ideas transparently documented |
| 10-14 21:30 | Round 1 v3 main chain: RAFT 40,000 tracks / STORM v4 10,000 years through the same H08 chain, 19 stations vs. ASCE 7-22 (public GIS, V700 19/19, V50 18/19 with KEYW missing) | Gate 0 passed (STORM V700 median −8.87%); Gate 1 failed (RAFT median absolute bias 30.40% > 15%, and > 1.1×STORM 9.21%); RAFT biased low across all four segments by −26% to −40%, not a single-station artifact; conversions 1.03–1.12, POCI ±5, and annual rates 10–20 do not change the verdict; RAFT V700 tail is supported by only ~3 storms; SPIN third chain (borrowing RAFT Pc/RMW formulas) V700 46.5–64.8 for reporting only; ~8.7 CU. Proposed to shelve, pending review (6 reports) |
| 10-14 | Grok quick check (issued by Node B) | Existing design-wind studies all use hybrid methods; free-running AI typhoon intensities are generally biased weak; RAFT data are public (Zenodo 10392725, CC BY 4.0); 8/8 DOIs exist. Clues only |

# Parameter Source Table

(Wind field chain parameters carry over from the B1+B4 page parameter source table; this page lists only parameters newly added for B2) | Parameter | Value | Source or "Assumption" | Places Checked | | --- | --- | --- | --- | | STORM simulation years | 10,000 years (10 files of 1,000 years per basin; North Atlantic Poisson λ=10.8/yr, KHFQRECZ PDF p. 4; "14.5/yr" was unfounded and has been corrected) | Bloemendaal et al. 2020 wind speed probability paper PDF p. 8 (Zotero UKPE6ZD3) | Pending ideas verification of dataset documentation | | STORM wind averaging time | 10 m 10 min mean sustained wind | Same paper, PDF pp. 2, 8 | 10 min→1 min conversion factor to be checked (check literature before assuming) | | RAFT equivalent years / annual rate | 12.075 storms/yr (3,312.6 equivalent years), sensitivity 10/14.5/20 | Assumption (observational proxy): HURDAT2 2026-09-23 release, 1979–2018 TS/HU ≥35 kt and tropical lifetime ≥24 h, matching the 33 kt genesis threshold and <1 day exclusion in Xu 2024 PDF pp. 3–4 (PREREG_v6) | Xu 2024 PDF pp. 3–5, 9, Zenodo fields, | | RAFT wind averaging time, RMW source, environmental pressure | To be checked | — | Same as above | | STORM/RAFT environmental pressure POCI | 1010 hPa, sensitivity 1005/1015 | Assumption (track files in both catalogs lack this field, which H08 requires) | Both original papers, Zenodo/4TU fields, | | Output convention | 10 m, z0=0.03

m, 3 s gust | Vickery et al. 2010 (Zotero GPEGNRXQ) PDF p. 8 | Exact ASCE 7-22 original text page numbers pending | | 1 min→3 s gust factor (open terrain) | Over-water 1 min → open-terrain 3 s combined multiplier 1.07, sensitivity 1.03–1.12 (PREREG_v1) | Primary candidate: Simiu et al. card 2E23NHAE (10 m open water 1 min ↔ open terrain 3 s, PDF pp. 1–3, Table 1; EVA-STEP3-C07 restricts use to matching conventions) | ASCE 7-22 commentary Durst curve, gust factor card ZR7S3SEJ | | Over-water baseline → open-land conversion | Retains v1 formula (500 m reference logarithmic profile, land z0=0.03 m) | B1+B4 (assumption, range 200–1000 m) | Same as B1+B4 | | ASCE 7-22 station values | To be retrieved | Public sources (ATC Hazards by Location, etc.), without using Liang Hu's credentials | — |

## Open Items and Next Steps

- **21:50 Annual-rate version returned**: Using the HURDAT2 observational proxy rate (12.075 storms/yr) as the primary gating version, RAFT is at 30.95% and still fails the gate, leaving the conclusion unchanged; the evidence package reflects the 14.5 version, and v6 will be verified together during the verdict.
- **21:30 Round 1 results returned**: Gate 0 passed, Gate 1 failed, proposed to shelve; per protocol, abandoning a route requires review first, and `` ` `` (count 6) is running, with status to be finalized after the verdict. The annual rate still used 14.5/yr (not yet updated to HURDAT2 as requested at 19:54), but the 10–20/yr sensitivity shifts V700 by only −3.4%/+1.5%, which cannot close a 30% gap; submitted to reviewers for confirmation.
- 20:35 check: Supplement reported back—Supplements 1 and 2 were already incorporated into the existing plan, so no duplicate runs were launched; **ASCE 7-22 V700 values have been retrieved from public GIS for all 19 stations** (≥12 stations, so Gate 0 / Gate 1 are judged as usual without falling back to relative comparison). Before starting the main run, ideas caught an RMW unit bug during self-check (CLIMADA `tctrack_to_si` expects nautical miles, whereas the original script passed meters, causing double conversion), wrote PREREG_v3 (20:06) voiding the first 6 batches, and reran in ; simultaneously changed the handling of Pc>POCI to follow CLIMADA's default clipping (v1 originally stated "discard"). All changes occurred before any station statistics or gate evaluations, with zero parameter tuning. Separately wrote PREREG_v4_SPIN (third chain for reporting only, borrowing RAFT's empirical Pc/RMW formulas and assuming POCI = 1010). B2 Lead re-verified the B1+B4 chain: B-deck RMW is natively in nautical miles and passed directly, so **B1+B4 results are unaffected by this bug**.
- Note: The file named PREREG_v3 by ideas is the unit fix, not the change of RAFT annual rate to HURDAT2 requested by B2 Lead at 19:54 ( still queued, to become the next PREREG version); that decision preceded any results, and because annual rate only scales empirical return periods, it can be updated in post-processing, with the HURDAT2 annual-rate version serving as the authoritative gating version.
- 19:54 check: ideas has started work; PREREG_v1 (19:50) and v2 (switching STORM to the deduplicated v4 release) have been written, and no station wind speeds have been computed yet. B2 Lead has instructed that v3 be issued prior to computation: set the primary RAFT annual occurrence rate to the contemporaneous, matched-threshold HURDAT2 observed annual mean, and downgrade STORM's 14.5 storms/yr to a sensitivity case

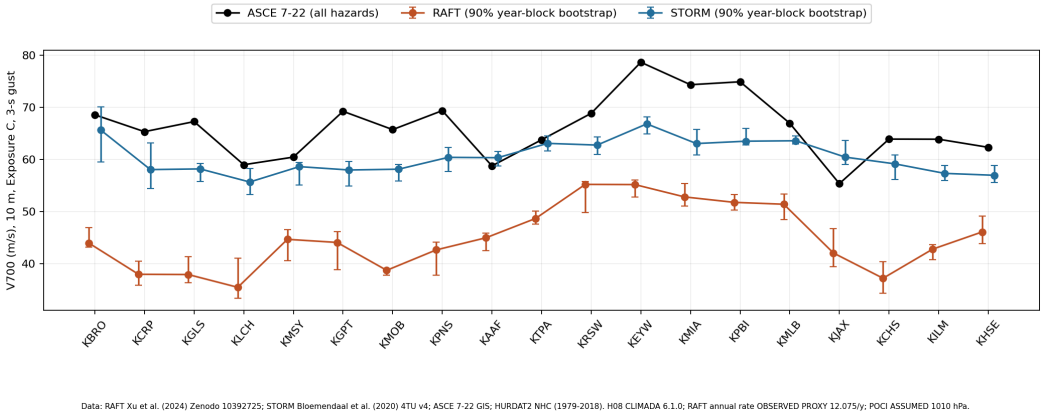
- (Completed) Round 1 and annual-rate version, now shelved
- (Completed 10-14) 7 litwiki cards created; Bose 2023 final publication page numbers not yet verified, so its page numbers are not cited for now
- Deep Research Step 2: dispatch once quota resets (`), conclusions

# Literature Basis

- 2026-10-16 data: 7,524 NOAA HRD dropsonde 10–300 m typhoon wind profiles (Ma et al. 2026, Zenodo 10.5281/zenodo.21868265, CC BY 4.0; approved X2 by Liang Hu with instructions to archive) → (backed up on CRC). Useful for profile shape, 10–300 m wind shear, comparison against H08/empirical profiles (over ocean), and profile constraints if B1+B4 is restarted; not overland profiles. Does not alter this node's status or gates.
- Cards (created in litwiki on 10-14, index), all located in :
- RAFT `NR_xu_2024_raft.mdyear` denotes the environmental year (PDF pp. 3, 5); intensity is 10 m 1 min, Pc and Rmax are given by empirical formulas, and no environmental pressure is provided (PDF p. 4 Eqs. 2–3); the authors self-report that across 51 US coastal cities, landfall 99th-percentile wind is ~8 kt low and Rmax is ~19% low on average (PDF pp. 6–7)—pointing in the expected direction of a low bias for the RAFT chain.
- STORM generation paper , wind speed probability paper.
- Vickery 2010 : 10 m open-terrain peak gust, Category II = 700 years.
- SPIN : full synthetic track set is publicly available, Zenodo 10.5281/zenodo.20795548, CC BY 4.0 (PDF p. 19 / journal p. 7321) → usable as a third chain for reporting only.
- Bose 2023 : catalog available only "upon reasonable request," not public; final PDF page numbers unverified.
- Snaiki & Wu 2019.
- Grok quick-check report (clues, unreviewed) see table above
- ASCE 7-22:
- Vickery 2000 track model:

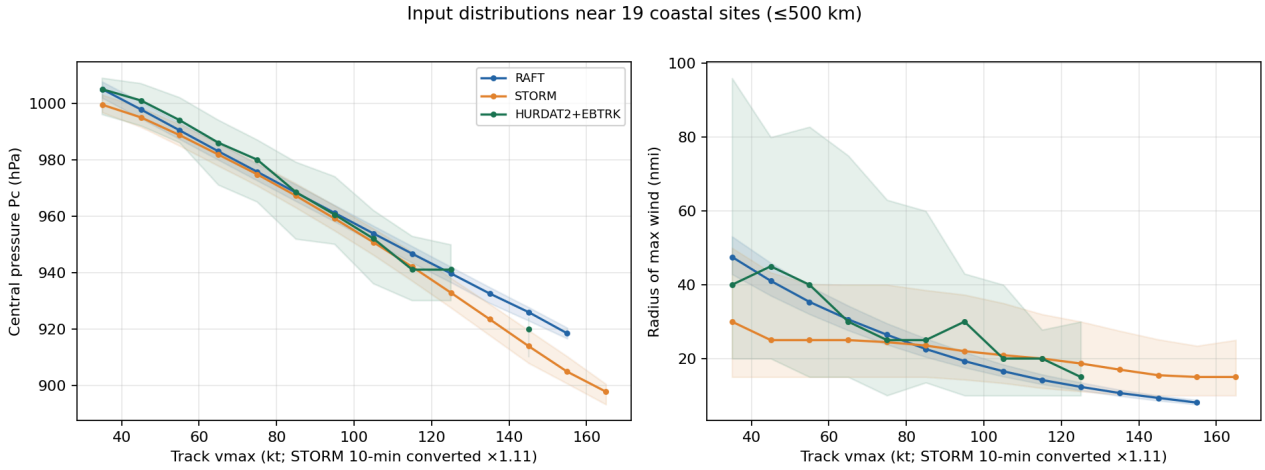
# Liang Hu Notes

## Key figures



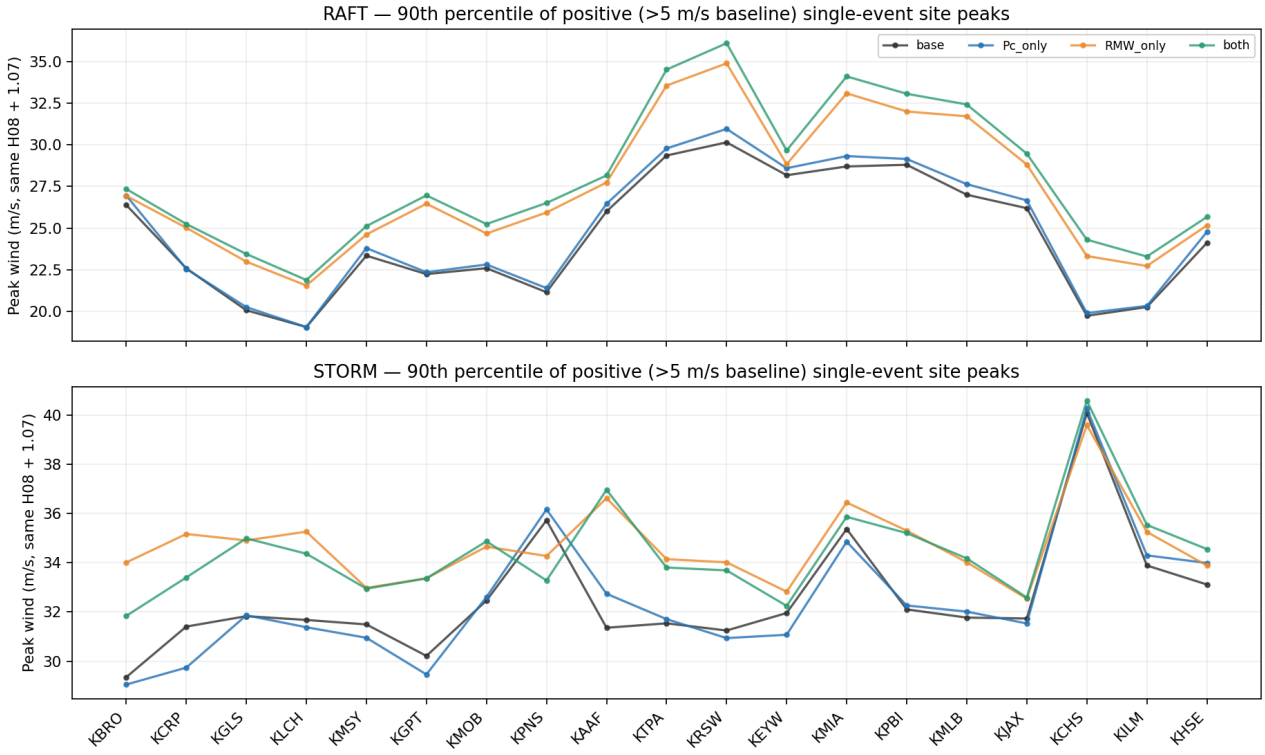

Data: RAFT Xu et al. (2024) Zenodo 10392725; STORM Bloemendaal et al. (2020) 4TU v4; ASCE 7-22 GIS; HURDAT2 NHC (1979-2018). H08 CLIMADA 6.1.0; RAFT annual rate OBSERVED PROXY 12.075/y; POCI ASSUMED 1010 hPa.

v700 comparison



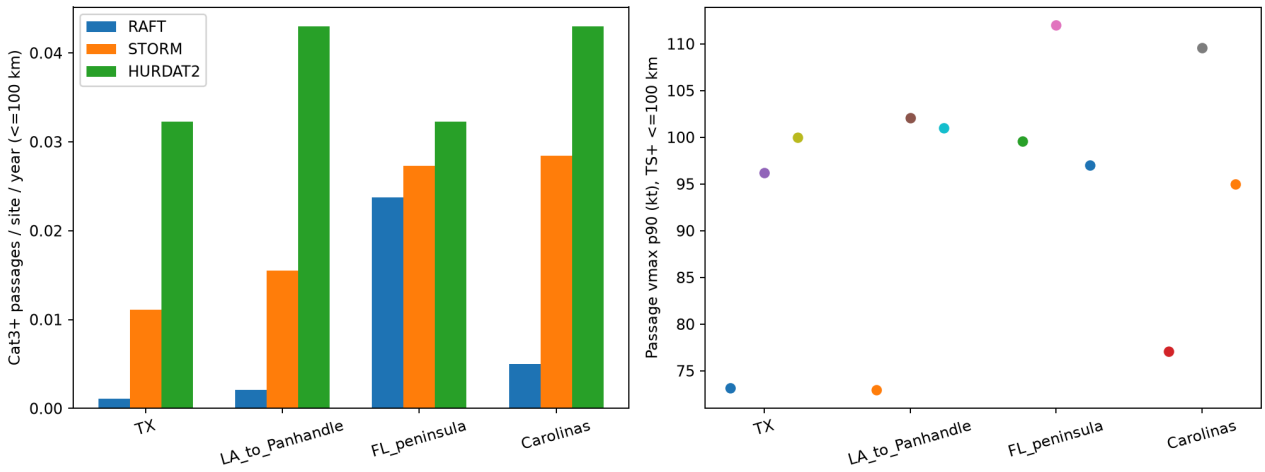

Sources: Xu et al. 2024 RAFT; Bloemendaal et al. 2020 STORM v4; NHC HURDAT2 2026 + CIRA EBTRK v3.0.2. Native track nodes; shading P10–P90.

inputs comparison



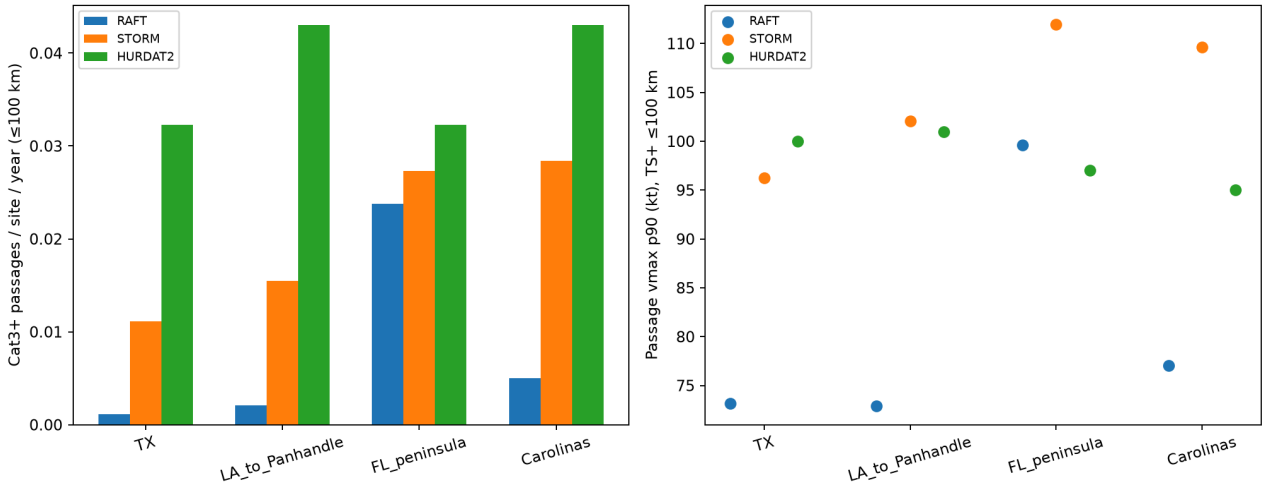

Sources: RAFT (Xu 2024), STORM v4 (Bloemendaal 2020), HURDAT2 + EBTRK; CLIMADA 6.1.0 H08. Deterministic 5% event sample; NOT return levels.

paired peak 5pct




Sources: RAFT Xu 2024; STORM v4 4TU; HURDAT2 NHC 2026; EBTRK CIRA. Native nodes, no hazard-model winds.

passage v1




Sources: RAFT Xu 2024; STORM v4 4TU; HURDAT2 NHC 2026; EBTRK CIRA. Native nodes, not model winds.

passage v1 legend